3.1.1 Atomic structure: Sheet 6

Electron configuration and ionisation energy · 16 marks · about 25 minutes

Key information

- Ionisation energy: energy needed to remove one mole of electrons from one mole of gaseous atoms (or ions).
- Successive IE equations always show gaseous ions losing one electron, eg $Ca^{+}(g) \rightarrow Ca^{2+}(g) + e^{-}$
- Fill sub-levels 1s 2s 2p 3s 3p 4s 3d 4p. For transition metal ions, remove 4s electrons before 3d.
- Across a period: IE increases (more protons, same shielding). Down a group: IE decreases (outer electron further away, more shielding).
- Dips: Group 3 (electron in higher-energy p sub-level) and Group 6 (paired p electrons repel).

Section A: Self-quiz

01 Give the full electron configuration for the calcium ion, Ca^{2+}

[1 mark]

02 Give the full electron configuration of a copper(II) ion.

[1 mark]

03 Which element has the highest first ionisation energy?

[1 mark]

A Aluminium
B Phosphorus
C Silicon
D Sulfur

04 Which statement about ionisation energies in Period 3 is not correct?

[1 mark]

A The first ionisation energy of Mg is greater than that of Na
B The first ionisation energy of Al is less than that of Mg
C The first ionisation energy of P is less than that of S
D Ar has the highest first ionisation energy in the period.

05 Which element has the highest third ionisation energy?

[1 mark]

A Li
B Be
C K
D Ca

Section B: Exam questions

06 This question is about ionisation energies.

06.1 Explain why beryllium is the element with the highest first ionisation energy in Group 2.

[2 marks]

06.2 Give an equation, including state symbols, to show the process that occurs when the second ionisation energy of beryllium is measured.

[1 mark]

06.3 Predict which element in Period 3 has the lowest second ionisation energy.

[1 mark]

07 This question is about s-block metals.

07.1 Explain why the second ionisation energy of calcium is lower than the second ionisation energy of potassium.

[2 marks]

07.2 Identify the s-block metal that has the highest first ionisation energy.

[1 mark]

08 Write an equation, including state symbols, to represent the process that occurs when the third ionisation energy of manganese is measured.

[1 mark]

09 Explain why the second ionisation energy of calcium is greater than the first ionisation energy of calcium.

[1 mark]

10 An atom of element R contains seven electrons.

10.1 Deduce the full electron configuration of an atom of element R.

[1 mark]

10.2 Identify R and deduce the formula of the compound formed when R reacts with the Group 2 metal in the same period as R.

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ (4s^0)$	Source: 7405/1 Jun 2018 Q04.1	1
02	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$	Source: 7405/1 Jun 2019 Q06.4 Do not allow $[Ar]3d^9$	1
03	**B**	Source: 7404/1 Jun 2018 Q18	1
04	**C**	Source: 7405/3 Jun 2025 Q16	1
05	**B**	Source: 7405/3 Jun 2024 Q07	1
06.1	M1 smallest atom / atomic radius OR smallest number of shells / energy levels OR outer electron is closest to the nucleus OR least shielding M2 strongest attraction between outer / valence electron(s) and nucleus	Source: 7405/1 Jun 2025 Q02.1–02.3 Allow comparative and superlative only for M1 and M2 Allow correct trends up or down Group 2	2
06.2	$Be^+(g) \rightarrow Be^{2+}(g) + e^-$ OR $Be^+(g) + e^- \rightarrow Be^{2+}(g) + 2e^-$		1
06.3	Mg / magnesium		1
07.1	M1 In $Ca^{(+)}$ (outer) electron(s) is further from nucleus OR $Ca^{(+)}$ loses electron from a higher (energy) orbital / 4(s) orbital / 4th energy level and $K^{(+)}$ loses electron from a 3(p) orbital / 3rd energy level M2 More shielding (in Ca^+)	Source: 7405/1 Jun 2018 Q04.2–04.3 Must be comparative Allow converse arguments	2
07.2	Be / beryllium		1
08	$Mn^{2+}(g) \rightarrow Mn^{3+}(g) + e^-$	Source: 7404/1 Jun 2017 Q01.2 Allow $Mn^{2+}(g) - e^- \rightarrow Mn^{3+}(g)$ State symbols are required; negative charge needed on electron	1
09	Electron removed from a positive charge / ion OR electron removed from smaller ion / closer to nucleus OR stronger attraction between same number of protons and fewer electrons	Source: 7405/1 Jun 2023 Q05.6	1
10.1	$1s^2\ 2s^2\ 2p^3$	Source: 7404/1 Jun 2018 Q01.2–01.3 Ignore commas, capitals and subscripts Allow $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$	1
10.2	(R is N (nitrogen)) Formula Be_3N_2	Accept Be_3R_2 only if stated R = nitrogen Accept N_2Be_3	1